BBF202E Numerical Methods in Computer Engineering

Spring 2025-2026 Term Project

Topic: Robust Spatio-Temporal Data Imputation and Nowcasting in Agricultural Sensor Networks via Iterative SVD

Due Date: May 17, 2026, 23:59

Honor Code: By turning in this assignment, I agree by the ITU honor code and declare that all of this is my own work. Any form of academic dishonesty, including code sharing or plagiarism, will be prosecuted according to the university's disciplinary regulations.

1 Important Notes and Submission Policy

- **Submission Platform:** All materials must be uploaded via **Ninova**. Submissions through e-mail or late entries will **not** be evaluated.
- **Deliverables:** You must submit a single `.zip` file titled `StudentID_Project.zip` containing:
  1. A well-documented and commented Jupyter Notebook (`.ipynb`).
  2. A formal project report in **PDF** format (Maximum 5 pages, provided IEEE Template).
- **Format:** The code must be clean, structured, and **well-documented**. Every non-trivial code block must include comments explaining the numerical logic behind it.
- **Originality:** All submissions will be screened by plagiarism detection tools.
- **Contact:** If you have any questions, please contact Res. Asst. Mehmet Selahaddin Şentop via `selahaddin.sentop@itu.edu.tr`.

2 Introduction and Problem Background

2.1 The Significance of Missing Data in Sensor Networks

Environmental monitoring systems, such as the **TARBİL** network in Turkey, provide critical real-time data for agricultural meteorology, irrigation management, and climate change modeling. These systems rely on a vast array of sensors measuring variables like temperature, humidity, solar radiation, and wind speed. However, real-world sensor deployments are notoriously prone to data discontinuities.

Data gaps can occur due to:

- **Hardware Failures:** Sensor degradation or physical damage to the equipment.
- **Energy Depletion:** Power outages in remote solar-powered stations.
- **Communication Latency:** Signal loss during data transmission in rural areas.

The presence of missing values (gaps) is not just a database issue; it is a fundamental obstacle for prediction models. Most machine learning and numerical forecasting models require complete data matrices. Inaccurate imputation leads to "cascading errors," where a small "guess" in the missing data results in massive failures in agricultural yield prediction or flood warnings.

2.2 Spatio-Temporal Correlations and Low-Rank Structures

As established in recent literature [1], agricultural data is characterized by **spatio-temporal dependencies**. This means:

1. **Temporal Correlation:** The temperature at 10:00 AM is highly likely to be related to the temperature at 11:00 AM.
2. **Spatial Correlation:** Stations located in the same region (e.g., three nearby TARBİL stations) will experience similar meteorological patterns.

Traditional methods like **Mean Imputation** or **Linear Interpolation** fail because they treat each variable or each time-step in isolation. More complex methods like **MICE** or **K-Nearest Neighbors (KNN)** capture some relationships but often struggle with the non-linear and high-dimensional nature of sensor data. Singular Value Decomposition (SVD) offers a superior numerical framework by uncovering the "latent structure" of the data. By treating the sensor network as a single matrix, SVD identifies the most dominant patterns (singular values) that govern the entire system, allowing us to reconstruct missing information with high fidelity.

3 Detailed Technical Tasks

3.1 Task 0: Mandatory Data Interface and Feature Selection

To ensure compatibility with the **Blind Test Evaluation**, all students must adhere to the following data structure. Any deviation that causes the grading script to fail will result in a zero grade for the implementation section.

- **Feature Selection:** You must use exactly the following four features from each station: `sicaklik_2m_mean`, `nispi_nem_2m_mean`, `gunes_radyasyon`, and `ruzgar_hizi`.

- **Matrix Construction:** Your final input matrix $X \in \mathbb{R}^{N\times 12}$ must be constructed by concatenating these 4 features from Station 1, Station 2, and Station 3 **side-by-side (horizontally)** in that specific order. This results in exactly 12 columns (4 features $\times$ 3 stations = 12).

- **Generalization:** Your code must dynamically handle the number of rows (N), as the test set may cover a different time interval than the provided development set. Do not hard-code any row indices.

- **Missing Values (NaNs):** You will find that the raw data already contains some missing entries. Before starting your experiments, you should perform a preliminary cleaning (e.g., dropping rows where all stations are missing simultaneously) to have a "base" ground truth.

3.2 Task 1: Data Integration and Generalization (The "Blind Test" Warning)

You are provided with three files: `station1.csv`, `station2.csv`, and `station3.csv`. You can use any pieces of them and play with them to build your optimal SVD-based data imputation model based on the data type and length of missingness.

- **Preprocessing:** Align all data by the `zaman_olcum` column. Handle any duplicate or mismatched timestamps.

- **Standardization:** You **must** standardize each column to zero mean and unit variance. Without this, variables with larger magnitudes (e.g., Solar Radiation) will dominate the singular values, rendering the reconstruction of smaller-scale variables (e.g., Temperature) impossible.

- **Generalization Requirement: CRITICAL:** While you develop your code using these three files, your algorithm must be **agnostic to station IDs and time intervals**. During grading, your notebook will be tested with a **Blind Test Set** (different stations/dates but the same column format). If your code contains hard-coded station names or assumes a specific row count, it will fail the test.

3.3 Task 2: SVD Solver from Scratch

The use of `numpy.linalg.svd` or any other library-based decomposition is **strictly prohibited**. You must implement a function `my_svd(Matrix)` using:

1. **Eigenvalue Decomposition:** Implement the **Power Method with Deflation** or the **QR Algorithm** to find eigenvalues of the covariance-like matrix.

2. **Orthogonality Verification:** In your notebook, provide a check to prove that $U^T U \approx I$ and $V^T V \approx I$.

3.4 Task 3: Iterative Singular Value Thresholding (SVT)[2]

Implement the iterative matrix completion algorithm:

1. **Step A:** Fill the missing entries Ω with **zero**. Since the data is standardized in Phase 1, zero represents the mean of the distribution, providing a neutral starting point for the iteration.

2. **Step B:** Decompose the matrix and apply a **low-rank truncation** by keeping only the top r singular values.

3. **Step C:** Update **only** the missing entries with values from the reconstructed matrix.

4. **Step D:** Repeat the process until the Frobenius norm of the difference between consecutive reconstructions is minimized: $\|X_{k+1} - X_k\|_F < \epsilon$. We recommend a threshold of $\epsilon = 10^{-4}$ for numerical stability.

3.5 Task 4: The "Stress Test" Experiments (Mandatory)

Do not simply report a single RMSE. You must perform and document the following:

1. **The Feature-Wise Success:** Compute RMSE separately for *Temperature* and *Wind Speed*. In your report, discuss why SVD might be more "confident" in reconstructing temperature (smooth dynamics) compared to the stochastic nature of wind speed.

2. **The Gap Duration Analysis:**

   - *Scenario A (Random):* 20% of the data is missing randomly for one station in a given time interval.
   - *Scenario B (Burst):* A continuous 24-hour block is missing for one station.
   - *Scenario C (Blackout):* A continuous 72-hour block is missing for one station.

   Compare the recovery performance across these scenarios.

4 Experimental Interrogation and Reporting

We expect you to "interrogate" your results through rigorous experimentation. In your **IEEE-formatted report**, you must address:

A. **Variable Analysis:** Compare the imputation success across different sensor types. Is SVD more successful in reconstructing *Temperature* compared to *Wind Speed*? Discuss the physical reasons behind this (e.g., smoothness vs. turbulence).

B. **Temporal Window Stress Test:** Perform experiments with different gap lengths (e.g., a 3-hour gap vs. a 72-hour continuous blackout). At what point does the spatio-temporal correlation break down?

C. **Rank Selection Analysis:** Provide plots of RMSE vs. Rank (r). Discuss how many "principal components" are actually needed to represent the agricultural dynamics of the region.

D. **Numerical Challenges:** Document the difficulties you faced, such as sign ambiguity in eigenvectors or convergence speed issues during the SVT process.

5 Evaluation Criteria

- **80% Implementation Quality:** Accuracy of scratch SVD (40%), Robustness against Blind Test (20%), Analytical Depth of Experiments (20%).
- **20% Project Report:** Adherence to IEEE format, clarity of documentation, and professional citation of sources.

References

[1] M. S. Şentop, M. Yücel and B. B. Üstündağ, "Spatio-Temporal Missing Data Reconstruction by Using Deep Neural Networks in Agricultural Monitoring Systems," *2023 11th International Conference on Agro-Geoinformatics (Agro-Geoinformatics)*, 2023. DOI: 10.1109/Agro-Geoinformatics59224.2023.10233578.

[2] J. F. Cai, E. J. Candès and Z. Shen, "A Singular Value Thresholding Algorithm for Matrix Completion," *SIAM Journal on Optimization*, 2010.